Reg. No. | | | | | | | | | | | [16CIC405]

FOURTH SEMESTER B.TECH EXAMINATION, JUNE – 2018

ANALYSIS AND DESIGN OF ALGORITHMS

Time:3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full questions. Choosing at least **TWO** questions from each part.
2. Any missing data may be suitably assumed

PART - A

1. a. Discuss the various stages of algorithm design and analysis process using flow chart. **07**

   b. List and explain the basic issues related to Algorithm. provide the Euclid's algorithm for computing GCD. **07**

2. a. Define the asymptotic notations used for best case average case and worst case analysis of algorithm. **06**

   b. Show that the function **08**

   $3n^2 - 100n + 6 = O(n^2)$

   $3n^2 - 100n + 6 = O(n^3)$

   $3n^2 - 100n + 6 \neq O(n)$

   $3n^2 - 100n + 6 = \Omega(n^2)$

   $3n^2 - 100n + 6 \neq \Omega(n^3)$

   $3n^2 - 100n + 6 = \Omega(n)$

   $3n^2 - 100n + 6 = \Theta(n^2)$

   $3n^2 - 100n + 6 \neq \Theta(n^3)$

   $3n^2 - 100n + 6 \neq \Theta(n)$

3. a. Write an algorithm for Brute-Force String Matching. Apply the algorithm for the given problem having main string **N O B O D Y _ N O T I C E D _ H I M** search for substring **NOT.** **09**

   b. With an example, show how an exhaustive search may be applied for travelling salesman problem. **05**

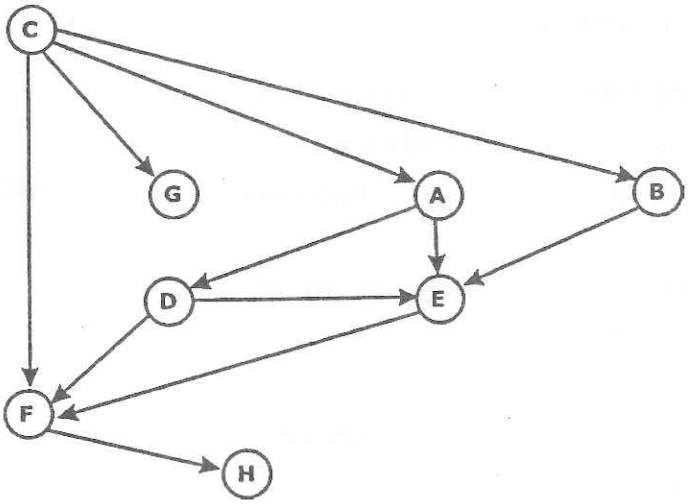
4. a. Discuss how quick sort works to sort an array and trace for the following data set. Draw the tree of recursive calls made. 85, 24, 63, 45, 17, 31, 96, 50. **08**

   b. Illustrate the operation of Radix sort on the following list of 3 digit numbers : 329,457,657,839,436,720,355 **06**

PART - B

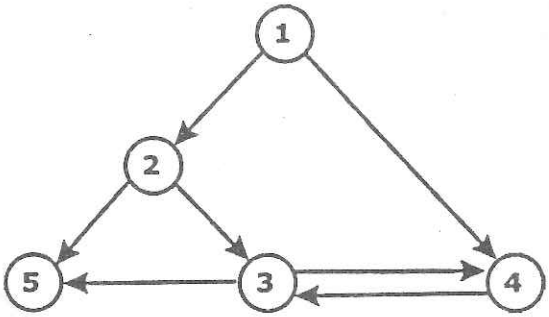
5. a. Give BFS and DFS algorithm. Clearly bring out the differences and comment on the complexity. **08**

b. solve the topological sorting problem for the following diagraph using DFS approach. **06**



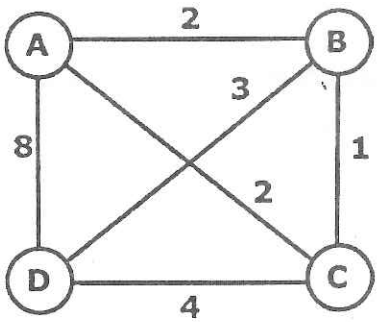
6. a. Explain the steps involved in dynamic programming. Apply the dynamic programming to compute Binomial Coefficient for n=8, k=4. **06**

b. Provide the rules and Generate the transitive closure of the graph given below using dynamic programming. **08**

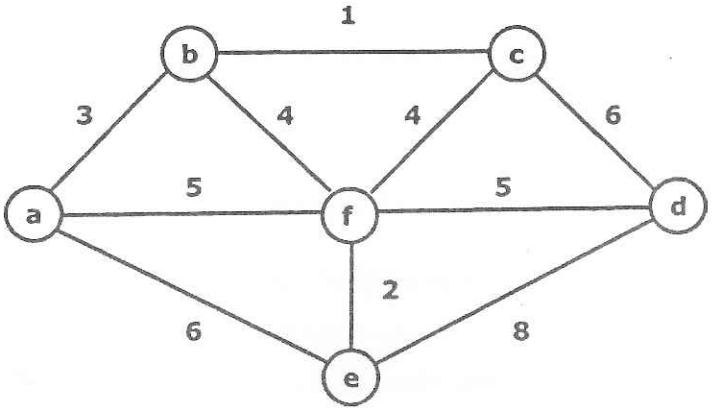
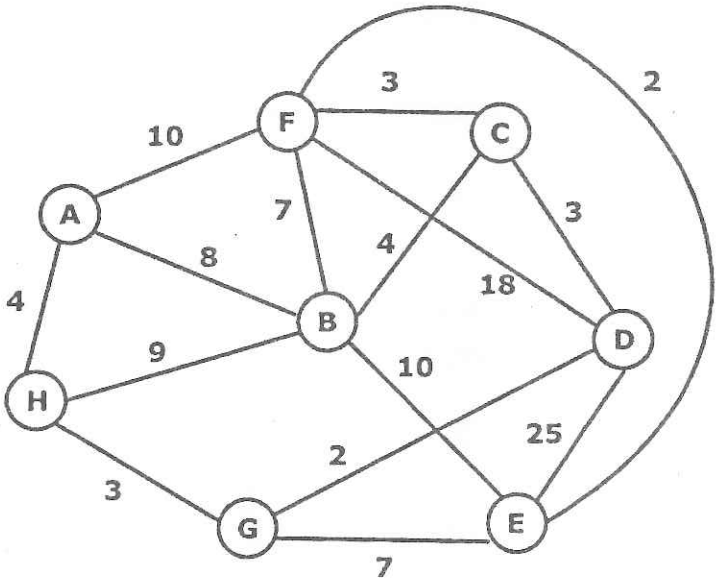


7. a. List the elements of greedy approach. **06**

b. Describe Spanning tree. For a given weighted graph generate the spanning using greedy approach. **08**



8. a. Brief the steps involved to generate MST using kruskal's algorithm and apply the algorithm for the following graph. **06**



b. Give Prim's algorithm to find the MST. Apply the algorithm for the given undirected graph to find minimum cost spanning tree from D. **08**